

Room-Conditioned Door Search as a Learning-Augmented Interface

Q Learning-Augmented Algorithms: Guarantees, Construction Mechanisms, and System-Level Implications [arXiv:2609.04787](#)

P Concept-Guided Exploration: Building Persistent, Actionable Scene Graphs [arXiv:2608.23650](#)

Research phase: PAUSE after 1 round, 0 ITERATE, 0 REVISE

The pair

Q surveys learning-augmented algorithms and requires an explicit prediction, error measure, objective, and benchmark before consistency or robustness can be claimed. P describes a robot whose estimated room walls constrain door detection, but reports geometric and topological experiments rather than a prediction-independent door-search guarantee [\[ledger 1, 2, 15\]](#).

Connection pursued

The question became whether P can save search work by using room walls while retaining visible door evidence when its room model is wrong. Q supplies a way to specify the required guarantee and warns that component guarantees do not compose without interface sensitivity and benchmark relations; it supplies no ready-made theorem for P [\[ledger 3, 14, 15\]](#).

What was found

P's door agent activates only after a current room is established. Its detector first retains LiDAR points within 0.15 m of the estimated walls, then applies range-derivative, edge-pair, width, and wall-membership tests. Accepted room models currently remain fixed even if later evidence contradicts them [\[ledger 3, 11, 15\]](#).

This creates a conditional obstruction. Suppose a wrong, persistent room model excludes the evidence for a visible door on every visit, while a prediction-independent full-field detector would find it in finite time. The gated detector never examines that evidence, so its discovery latency is infinite relative to that comparator. Consequently, no finite prediction-independent discovery-latency ratio follows from P's current architecture under these assumptions [\[ledger 5, 15\]](#).

A limited positive statement follows from geometry. Let W^* be the true walls, $\widehat{W}$ the estimated walls, and p a relevant door point. If

$$\sup_{w \in W^*} \text{dist}(w, \widehat{W}) \leq \epsilon \quad \text{and} \quad \text{dist}(p, W^*) \leq b,$$

then the triangle inequality gives

$$\text{dist}(p, \widehat{W}) \leq b + \epsilon.$$

A gate of width at least $b + \epsilon$ therefore retains p . This is a candidate-retention result, not a door-recall result: every later detection test needs its own completeness or margin assumption [ledger 4, 9, 11, 15].

Objections and resolution

The hard gate can change its retained set abruptly at the threshold, so Q’s Lipschitz error-propagation condition cannot be invoked without an additional margin or a revised gate [ledger 9, 15]. P’s reported mean room-dimension error cannot serve as the required wall-offset bound, because the quantities differ [ledger 11].

Periodic full-field audits are a possible fallback only under stated sensing assumptions. If a complete audit occurs within every window of τ scans and a door stays visible throughout such a window, discovery takes at most that window. A door visible for only one unaudited scan defeats this argument unless raw observations can be replayed under additional storage and completeness assumptions [ledger 12, 18, 20]. None of these statements establishes a navigation-level competitive ratio [ledger 20].

Outside sources and remaining work

The peers checked Eberle et al., *Robustification of Online Graph Exploration Methods* (<https://arxiv.org/abs/2112.05422>), which already studies learning-augmented exploration, and Xu et al., *Deep GrabCut for Object Selection* (<https://arxiv.org/abs/1707.00243>), which gives an analogous reason to soften a hard predicted region. These checks limit any novelty claim: the prospective contribution is specific to P’s geometric gate and downstream detection pipeline [ledger 7, 10, 13, 15].

The evidence is thin on the central feasibility question. It remains unknown whether P can obtain a sound online wall-error certificate, whether a full-field detector can be complete under realistic sensing, and whether the resulting search-work and detection-delay trade-off improves relevant prior methods [ledger 14, 15, 20]. The smallest next step is to specify a visible-door test model and a complete full-field comparator, then perturb accepted room walls in P’s detector and measure which doors survive each stage of the pipeline. That experiment would determine whether a useful margin or audit theorem is worth pursuing [ledger 14, 15].